

ARPON KAPURIA

✉ arpkapuriah@gmail.com | 🌐 arponkapuria.github.io | 📞 +8801797288296 | 🌍 Bangladesh

EDUCATION

National Institute of Technology Trichy

Bachelor of Technology (Computer Science and Engineering)

Tamil Nadu, India

November 2020 – June 2024

SKILLS

- **Programming Languages:** Python, C++, SQL
- **Backend & Data:** FastAPI, PostgreSQL, Redis, Qdrant, Docling
- **AI / ML:** PyTorch, scikit-learn, Hugging Face, Unsloth, LangChain, LangGraph, RAGAS, vLLM
- **Infrastructure & Ops:** MLflow, DVC, Langfuse, Docker, Kubernetes, AWS
- **Concepts:** Fine-tuning, Prompt Engineering, RAG, AI Agents, LLM Inference/Serving, Evaluation

WORK EXPERIENCE

Indian Institute of Technology Bombay

Summer Research Intern | MeDAL Lab

Mumbai, India

May 2023 – July 2023

- Reproduced self-supervised learning frameworks BYOL/SimCLR in PyTorch and experimented with a custom hybrid contrastive-BYOL loss on STL10, where explicit negatives raised BYOL kNN accuracy from **33%** to **60%** and reduced dimensional collapse.
- Implemented an automated DICOM anonymization pipeline (Pydicom) to remove patient metadata and used CRAFT text detection to mask burned-in PHI in X-ray images.

National Institute of Technology Trichy

Research Assistant (Software) | Industrial Automation Lab

Tamil Nadu, India

October 2022 – February 2023

- Developed Flutter frontend of a cross-platform smart-home-automation app for energy monitoring, switch control, and UPI-based digital billing in a solar-powered microgrid serving 20 households in Assam, as part of a govt. funded project.

PROJECTS

DocRAG: Cost-Aware Multi-Tenant Agentic RAG Pipeline | GitHub

Python, FastAPI, LangGraph, PostgreSQL, Qdrant, Redis Stack, Docling, FastEmbed, Langfuse, Docker

- Designed a fault-tolerant RAG pipeline for grounded multi-turn Q&A over user documents, from async ingestion to serving, running within free-tier LLM quotas. Integrated agent orchestration with a query planner for tool selection, a two-tier semantic cache and multi-provider LLM routing with circuit breakers and failover for cost-efficient availability (**zero failures** in a forced-outage test).
- Enforced multi-tenant isolation in a login-free app: server-issued sessions scope access to every store (vector, relational, object, cache), with cascade deletion on expiry, verified against cross-session leakage. Added inbound rate limiting and regex guardrails against prompt injection and prompt leakage, tested on both attack and legitimate queries.
- Added tracing and wrote an evaluation harness in RAGAS on a hand-verified golden dataset and compared five retrieval setups. Hybrid retrieval with reranking improved Hit@5 by **17%** over semantic-only search and reached **95%** context recall, and the full citation-grounded pipeline raised answer correctness by **9%** over the same model without retrieval, with a **100%** correct refusal rate.

NanoServe: A Tiny LLM Inference Engine | GitHub

Python, PyTorch (MPS), Hugging Face Transformers, Qwen3-1.7B

- Built a tiny LLM inference engine for Apple Silicon on top of Hugging Face Transformers to study serving optimizations. Implemented a paged KV cache with a block allocator in place of per-request max-length reservation, cutting internal fragmentation from **45–92%** to under **5%** and increasing pool capacity from **4 to 36** sequences; continuous batching then raised throughput **32%**.
- Improved latency under load with admission control (bounded queue, timeouts), chunked prefill (max inter-token latency **–49%**) and radix-tree prefix caching (TTFT **40%** lower on shared-prompts).

Last updated in October 2026